

Writeup for Blockhouse Quant. Research Intern Trial

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Taks 1: the Cross Impact of OFI in Equity Markets

What's the motivation behind measuring OFI at multiple depth levels of the order book?

Measuring OFI (use PCA) at multiple depth levels captures liquidity imbalances deeper in the book that are not visible at the best quote. Including multiple levels improves the explanatory power of OFI models for contemporaneous price changes across different stocks.

Why do the authors use Lasso regression rather than OLS for estimating cross-impact?

The authors adopt LASSO because the cross-impact model involves a large number of potentially highly correlated OFI regressors, and the ordinary least squares would overfit and yield unstable model.

Why is OFI considered a better predictor of short-term returns than trade volume?

OFI is better than raw trade volume as a predictor of very short-term returns because it directly measures the net signed buying versus selling pressure at the best quotes, capturing the immediate supply-demand imbalance that drives price moves.